Quinn M. Gallagher

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Education

Princeton University — Princeton, NJ
Ph.D. Candidate in Chemical and Biological Engineering — *Aug. 2022 – Present*
Advisor: Michael A. Webb

University of Pennsylvania — Philadelphia, PA
B.S.E. in Chemical and Biomolecular Engineering, *summa cum laude* — *Aug. 2018 – May 2022*

Honors and Awards

Hack Award for Water and the Environment, High Meadows Environmental Institute	2026
AI for Accelerating Invention Graduate Fellow, Princeton University	2025
William R. Schowalter Travel Grant, Princeton University	2025, 2026
Excellence in Teaching Award, Princeton University	2024
Francis Robbins Upton Fellowship, Princeton University	2022
NSF Graduate Research Fellowship, National Science Foundation	2022
Hugh Otto Wolf Memorial Prize, University of Pennsylvania	2022
Future Leader in Chemical Engineering, North Carolina State University	2021
Penn Engineering Exceptional Service Award, University of Pennsylvania	2021
Tau Beta Pi, University of Pennsylvania	2021
Manfred Altman Memorial Award, University of Pennsylvania	2020
Holtz Award, University of Pennsylvania	2019

Publications

8. Minimizing Data Requirements for Material Property Prediction via Cluster-based Training Set Selection
Q.M. Gallagher, S. Hasko, M.A. Webb, *ChemRxiv* (2026). Revisions submitted at *Nature Computational Science*. [Preprint]

7. A Local Structural Basis to Resolve Amorphous Ices
Q.M. Gallagher*, R.J. Szukalo*, N. Giovambattista, P.G. Debenedetti, M.A. Webb, *arXiv* (2026). Revisions submitted at *Nature Communications*. [Preprint] *Equal contribution

6. A User's Guide to Your First Self-Driving Liquid Handling Lab
A. Maroulis, D. Waynor, **Q.M. Gallagher**, R.A. Patel, M. Tamasi, D.C. Radford, M.A. Webb, A.J. Gormley, *Digital Discovery* (2026). [Link]

5. Pressure-Consistent Iterative Boltzmann Inversion for Coarse-Grained Molecular Dynamics
Z. Yu, R.J. Szukalo, **Q.M. Gallagher**, M.A. Webb, *Journal of Chemical Theory and Computation*, 21, 20 (2025). [Link]

4. Sustainable Recovery of Perfluoroalkyl Acids Using a Reusable Molecular Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, M.A. Webb, A. Criado, J. Mosquera, *Journal of Materials Chemistry A*, 13, 30 (2025). [Link]

3. Bayesian Optimization Hackathon for Chemistry and Materials
S. Baird, ..., **Q.M. Gallagher**, ..., Y. Zo, *ChemRxiv* (2025). [Preprint]

2. Data Efficiency of Classification Strategies for Chemical and Materials Design
Q.M. Gallagher, M.A. Webb, *Digital Discovery*, 4, 1 (2025). [Link]

1. Engineering a Surfactant Trap via Postassembly Modification of an Imine Cage
M. Pérez-Ferreiro, **Q.M. Gallagher**, A.B. León, M.A. Webb, A. Criado, J. Mosquera, *Chemistry of Materials*, 36, 18 (2024). [Link]

Teaching

Princeton University

Guest Lecturer for CBE 512: Machine Learning for Chemical Science and Engineering *October 2025*

Teaching Assistant for CBE 250: Separations in Chemical Engineering *Aug. 2024 – Dec. 2024*

Teaching Assistant for CBE 503: Advanced Thermodynamics *Aug. 2023 – Dec. 2023*
Received the Excellence in Teaching Award from Princeton's School of Engineering and Applied Science.

University of Pennsylvania

Teaching Assistant for CIS 160: Mathematical Foundations of Computer Science *Jan. 2020 – May 2022*

Teacher for Access Engineering *Jan. 2019 – May 2022*

Mentoring

Undergraduate Students:

- Khang Tran, AI Fluency Program (March 2026 –)
- Sonia Hasko, CBE (Sept. 2024 – May 2025)
- David Opong, CBE (Sept. 2023 – May 2024)

High School Students:

- Michelle Gao, Laboratory Learning Program (May 2024 – Aug. 2024)

Academic Service

Journal Reviewer: *ACS Macro Letters*

Professional Membership: AIChE, ACS

Outreach

Graduate Fellow, AI Lab, AI for Accelerating Invention Initiative, Princeton University *Jan. 2026 –*

Reviewer, NSF GRFP Workshop for CBE Students, Princeton University *Sept. 2022 –*

President, Access Engineering, University of Pennsylvania *Sept. 2018 – May 2022*

Vice President, Student Chapter of AIChE, University of Pennsylvania *Sept. 2018 – May 2022*

Secretary, Engineering Deans' Advisory Board, University of Pennsylvania *Sept. 2018 – May 2022*

Presentations

14. "Improving Machine Learning Extrapolation for Molecular Discovery" (*Poster*) **ChemAI NYC**, 2026.

13. "Improving Extrapolative Machine Learning for Molecular Design" (*Poster*) **NIST Artificial Intelligence for Materials Science Workshop**, 2026.

12. "Improving Extrapolative Molecular Property Prediction for Screening, Optimization, and Guided Generation" (*Poster*) **CECAM Flagship Workshop**, *Physics-aware Machine Learning for Molecules and Materials*, 2026.

11. "Minimizing Data Requirements for Material Property Prediction *via* Cluster-based Training Set Selection" (*Poster*) **Gordon Research Conference**, *AI for Materials, Energy and Chemical Sciences*, 2026.

10. "Automatic Construction of Data-Driven Collective Variables Using Local Atomic Environments" (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Machine Learning for Soft and Hard Materials II: Hard Matter and Methods*, 2025.

9. "Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships" (*Talk*) **AIChE Annual Meeting**, *CoMSEF: Automated Molecular and Materials Discovery: Integrating Machine Learning, Simulation, and Experiment*, 2025.

8. "Data-driven Order Parameters for Atomic Systems: A Case Study on Amorphous Ices" (*Talk*) **Princeton CBE Graduate Student Symposium**, 2025.

7. "An Interpretable Machine Learning Tool for the Classification of Atomic States in Molecular Simulations" (*Poster*) **6th Molecular Simulations Workshop (NJIT)**, 2025.

6. "Minimizing Data Requirements for Material Property Prediction" (*Poster*) **AutoML Center for Accelerated Formulations Engineering Workshop**, 2025.

5. "Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships" (*Poster*) **Accelerate Conference**, 2025.

4. "Maximizing the Data Efficiency of Experimental Planning Algorithms for Quantifying Material Structure-Property Relationships" (*Poster*) **New York City AI4Chemistry Summit**, 2025.

3. "Data-driven identification of collective variables for transitions between high-density amorphous and low-density amorphous ice" (*Talk*) **American Chemical Society, Middle Atlantic Regional Meeting**, 2025.

2. "Fewer Experiments, More Knowledge: Data-Efficient Machine Learning for Materials Design" (*Poster*) **Princeton CBE Graduate Student Symposium**, 2024.

1. "Modulating Heterochromatin Formation for Direct Cell Reprogramming" (*Talk*) **North Carolina State University, Future Leaders in Chemical Engineering National Award Symposium**, 2021.